Task 1: Load a Pre-trained LLM and Experiment with Prompt Engineering (30 points)

My code:

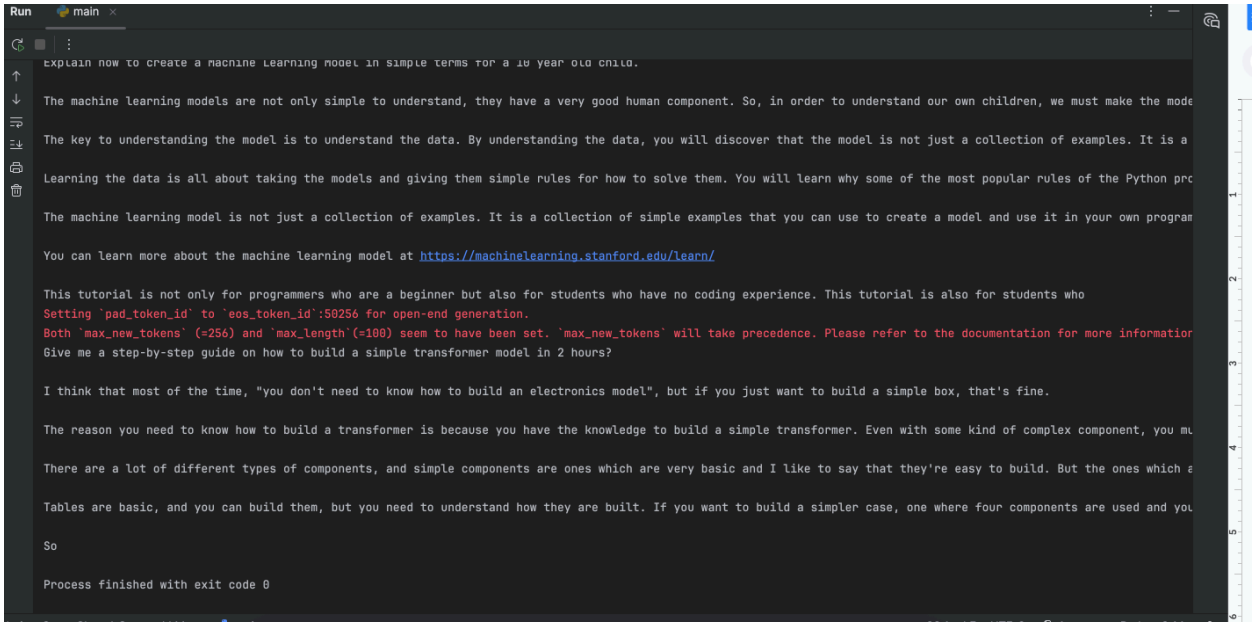
```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

# -- initial prompt

# prompts model
prompt = "Explain how to create a Machine Learning Model in
simple terms"
# defines starting sentence, max output length, # of answers
output = generator(prompt, max_length=100,
num_return_sequences=1,
# controls creativity (0.7 is medium), next word chosen from top
50 words
temperature = 0.7, top_k=50)
print(output[0]['generated_text'])

# -- prompt modification
# Casual Prompt

prompt = "What is a transformer model?"
output = generator(prompt, max_length=100,
num_return_sequences=1, temperature = 0.7, top_k=50)
print(output[0]['generated_text'])

# Detailed Prompt

prompt = "Explain how to create a Machine Learning Model in
simple terms for a 10 year old child."
output = generator(prompt, max_length=100,
num_return_sequences=1, temperature = 0.7, top_k=50)
print(output[0]['generated_text'])

# Structured Prompt

prompt = "Give me a step-by-step guide on how to build a simple
transformer model in 2 hours"
```

```
output = generator(prompt, max_length=100,
num_return_sequences=1, temperature = 0.7, top_k=50)
print(output[0]['generated_text'])
```

Output:

```
Run    main
/usr/local/bin/python3 /Users/nandana/CS450/Exploring-Open-Closed-Source-LLMs-/main.py
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰| 148/148 [00:00<00:00, 10202.60it/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Passing `generation_config` together with generation-related arguments=({'max_length', 'temperature', 'num_return_sequences', 'top_k'}) is deprecated and will be remove
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=100) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more informatior
Explain how to create a Machine Learning Model in simple terms

http://www.learnbmp.com/article/learn_machine_learning_model

The following example shows how we can create a machine learning model. The example uses an image representation tool called image-learning which is not part of the C#

public class ImageModel { public ImageSet<Direction> sets(Direction Direction); // An Image Set with a name and a function public ImageSet<Direction> images(ImageSet<Di

The image is constructed using the ImageSet<Direction> array of arrays and the ImageSet<Direction> array of functions.

public ImageSet<Direction> begin() { // Set the image to the specified Direction. SetSource(images); // Set the image to the specified function. set_size(images); // Se

Here's an image from the ImageSet class that we created. The image was created using a neural network and has the following function
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=100) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more informatior
What is a transformer model?

A transformer model is a system of values specified in a set of data that can be expressed in some form. In computer science, this means that data can be represented by

a->b->c ->d->e

where d is the data structure, c the data structure, and e the data structure.
```

```
Run    main
One might think that this can be interpreted as:

a->b->c ->d ->e ->c->d

which is exactly what it does. It writes the data structure. It stores the data structure. It performs the operation on it.

This is called a model transform. A model transformer is an abstraction of different data structures. Models are used to represent different things in your data structu

A model transformer is a data structure for objects. A model transformer is a data structure for the data.

A model transformer is not a data structure but a data structure that holds data. Model transformers are data structures that are applied to data and their values.

A model transformer is not a structure
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=100) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more informatior
Explain how to create a Machine Learning Model in simple terms for a 10 year old child.

The machine learning models are not only simple to understand, they have a very good human component. So, in order to understand our own children, we must make the mode

The key to understanding the model is to understand the data. By understanding the data, you will discover that the model is not just a collection of examples. It is a

Learning the data is all about taking the models and giving them simple rules for how to solve them. You will learn why some of the most popular rules of the Python pro

The machine learning model is not just a collection of examples. It is a collection of simple examples that you can use to create a model and use it in your own program

You can learn more about the machine learning model at https://machinelearning.stanford.edu/learn/

This tutorial is not only for programmers who are a beginner but also for students who have no coding experience. This tutorial is also for students who
```



Questions:

- The model's responses change based on the prompts I give it by changing its approach based on the specifications my prompt provides for it. In the first prompt I gave it, I wasn't very descriptive, asking it to explain how to create a machine learning model in simple terms. The model returned a very abstract response on creating a machine learning model which wasn't that simple and did not perform in the way I would have preferred, which was to give me a simple overview of a machine learning model. Similarly, when I prompted asking what a transformer model was, it returned an explanation of a model transformer, which is not the same thing. When I specified that the explanation should be so simple that a 10 year old child could understand it, the model returned a paragraph about education. Therefore, I understood that the model directly utilizes context clues from the prompt I provide it to return a response. When I specified the structured prompt as well, regarding the give me a step-by-step guide on building a simple transformer model in 2 hours, it wasn't really responding properly either, and rather going on a spiel about Table building.
- My findings for best prompt engineering is to be as specific as possible, without introducing new concepts that may diverge attention from your topic of interest. I also believe that decreasing the temperature helped me, as when I re-ran the program with the same questions and a decreased temperature of 0.6, I got better results from the model and better responses to my prompts. I found that the structured prompts helped, but some of them ended up confusing the model other than specifying, and the more general prompts were somewhat better received for the model in those cases. An example of this was when I prompted the model to explain the concept of a transformer model to me like I was a 10-year-old child and it began discussing education instead of anything relating to

my question. The better responses that I got were when my prompt wasn’t super specific but rather structured, allowing the model to follow a format yet still directly relate to the topic of interest rather than diverge into another topic accidentally.

Task 2: Use Hugging Face API or any other closed-source LLM API (e.g., OpenAI GPT, Claude) for Text Generation (20 points)

I utilized OpenAI's API Key (GPT-4o mini)

```python
# --- Task 2 --> OpenAI
from dotenv import load_dotenv
import os
from openai import OpenAI

load_dotenv()  # loads .env variables


client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))

response = client.chat.completions.create(
model="gpt-4o-mini",
messages=[{"role": "user", "content": "Explain Artificial Intelligence in simple terms."}],
)
print(response.choices[0].message.content)
```

[33] 3s 925ms

```
Artificial Intelligence, or AI, is a way for machines, like computers, to mimic human intelligence. This
means they can perform tasks that usually require human thinking, such as learning from experience,
solving problems, understanding language, and making decisions.

Think of AI as a smart assistant that can help with things like answering questions, recognizing faces in
photos, recommending movies you might like, or even driving a car. It does this by using data and
algorithms (a set of rules or instructions) to learn and improve over time, just like how we learn from
our experiences.
```

Questions:

- The output from the GPT 4.0 model was much better at recognizing the prompt and properly returning a good response compared to the local GPT-2. Immediately, the GPT 4.0 model provided a detailed and easy to understand response to my prompt, even including a detailed and helpful example to aid me in understanding the answer to my question that I prompted. On the other hand, the GPT-2 local model could not even comprehend my question well, and the majority of its responses were not directly answering my prompt. GPT-2 was an overall very unhelpful model, and the GPT-4 model performed much better, and was greatly helpful. Therefore, I would definitely utilize the 4.0 model rather than the locally running model for future tasks.

Task 3: Instruction Fine-Tune a Tiny Model on a Mini Dataset and Compare Outputs (100 points)

My toy_instructions.jsonl:


```
{"instruction":"Rewrite the sentence to be more polite.","input":"Give me
that.","output":"Could you please give me that?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Send me the
file.","output":"Could you please send me the file?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Open the
door.","output":"Could you please open the door?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Close the
window.","output":"Could you please close the window?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Finish the
report.","output":"Could you please finish the report?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Call me
back.","output":"Could you please call me back?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Explain this
problem.","output":"Could you please explain this problem?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Bring the
notebook.","output":"Could you please bring the notebook?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Turn on the
light.","output":"Could you please turn on the light?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Turn off the
fan.","output":"Could you please turn off the fan?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Pass the
salt.","output":"Could you please pass the salt?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Move this
box.","output":"Could you please move this box?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Help me with
this.","output":"Could you please help me with this?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Write this
down.","output":"Could you please write this down?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Check this
code.","output":"Could you please check this code?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Print this
page.","output":"Could you please print this page?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Bring some
water.","output":"Could you please bring some water?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Wait
here.","output":"Could you please wait here?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Read this
chapter.","output":"Could you please read this chapter?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Fix this
bug.","output":"Could you please fix this bug?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Stand
up.","output":"Could you please stand up?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Sit
down.","output":"Could you please sit down?"}
```

{"instruction":"Rewrite the sentence to be more polite.","input":"Send the email.","output":"Could you please send the email?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Complete the form.","output":"Could you please complete the form?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Hold this bag.","output":"Could you please hold this bag?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Take a seat.","output":"Could you please take a seat?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Wash the dishes.","output":"Could you please wash the dishes?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Lock the door.","output":"Could you please lock the door?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Unlock the car.","output":"Could you please unlock the car?"}
{"instruction":"Rewrite the sentence to be more polite.","input":"Bring your laptop.","output":"Could you please bring your laptop?"}

{"instruction":"Classify the sentiment as positive/negative.","input":"I love this movie.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"This food tastes amazing.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"I am very happy today.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"This product is excellent.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"The concert was fantastic.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"I really enjoyed this.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"The design looks beautiful.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Best experience ever.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"I like this app.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Wonderful performance.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"This is terrible.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Worst purchase ever.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"I hate this.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"This service is awful.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Very disappointing.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"The movie was boring.","output":"negative"}

{"instruction":"Classify the sentiment as positive/negative.","input":"This is the worst.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"I regret buying this.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Extremely annoying.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Poor quality.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Amazing work.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Horrible customer service.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Super fun experience.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Really bad taste.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Perfect result.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Completely useless.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Very enjoyable.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Totally frustrating.","output":"negative"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Absolutely loved it.","output":"positive"}
{"instruction":"Classify the sentiment as positive/negative.","input":"Very upsetting.","output":"negative"}

{"instruction":"Extract the city from the text.","input":"I flew from Seattle to Boston yesterday.","output":"Boston"}
{"instruction":"Extract the city from the text.","input":"She moved to Chicago last year.","output":"Chicago"}
{"instruction":"Extract the city from the text.","input":"He lives in Miami now.","output":"Miami"}
{"instruction":"Extract the city from the text.","input":"We stayed in Dallas for two days.","output":"Dallas"}
{"instruction":"Extract the city from the text.","input":"I traveled to Denver last week.","output":"Denver"}
{"instruction":"Extract the city from the text.","input":"They visited Austin during summer.","output":"Austin"}
{"instruction":"Extract the city from the text.","input":"She works in New York.","output":"New York"}
{"instruction":"Extract the city from the text.","input":"He moved to Portland recently.","output":"Portland"}
{"instruction":"Extract the city from the text.","input":"We drove to Phoenix.","output":"Phoenix"}
{"instruction":"Extract the city from the text.","input":"I studied in San Diego.","output":"San Diego"}

```
{"instruction":"Extract the city from the text.","input":"He relocated to
Houston.","output":"Houston"}
{"instruction":"Extract the city from the text.","input":"They met in San
Jose.","output":"San Jose"}
{"instruction":"Extract the city from the text.","input":"She moved to
Atlanta.","output":"Atlanta"}
{"instruction":"Extract the city from the text.","input":"I visited Los
Angeles.","output":"Los Angeles"}
{"instruction":"Extract the city from the text.","input":"He stayed in Las
Vegas.","output":"Las Vegas"}
{"instruction":"Extract the city from the text.","input":"They traveled to
Orlando.","output":"Orlando"}
{"instruction":"Extract the city from the text.","input":"We stopped in
Sacramento.","output":"Sacramento"}
{"instruction":"Extract the city from the text.","input":"She lives in
Tampa.","output":"Tampa"}
{"instruction":"Extract the city from the text.","input":"He studied in
Irvine.","output":"Irvine"}
{"instruction":"Extract the city from the text.","input":"I worked in
Oakland.","output":"Oakland"}
{"instruction":"Extract the city from the text.","input":"She visited
Seattle.","output":"Seattle"}
{"instruction":"Extract the city from the text.","input":"He moved to Long
Beach.","output":"Long Beach"}
{"instruction":"Extract the city from the text.","input":"We stayed in
Fresno.","output":"Fresno"}
{"instruction":"Extract the city from the text.","input":"I went to
Anaheim.","output":"Anaheim"}
{"instruction":"Extract the city from the text.","input":"They traveled to
Riverside.","output":"Riverside"}
{"instruction":"Extract the city from the text.","input":"She works in
Pasadena.","output":"Pasadena"}
{"instruction":"Extract the city from the text.","input":"He lives in Santa
Monica.","output":"Santa Monica"}
{"instruction":"Extract the city from the text.","input":"I drove to San
Francisco.","output":"San Francisco"}
{"instruction":"Extract the city from the text.","input":"They relocated to
San Diego.","output":"San Diego"}
{"instruction":"Extract the city from the text.","input":"We flew to
Boston.","output":"Boston"}
```


Training Loss Output Table:

Go to Cell 8 Python 3.

[5] 9m 44s

Loading weights: 100% 76/76 [00:00<00:00, 3941.87it/s]

[135/135 09:35, Epoch 3/3]

Step	Training Loss
5	3.836280
10	2.662162
15	2.009030
20	1.581615
25	1.345527
30	1.053076
35	1.107251
40	1.017458
45	0.919491
50	0.813166
55	0.728662
60	0.731694
65	0.576939
70	0.622448
75	0.609539
80	0.654780
85	0.646335
90	0.539000

Go to Cell 8 Python 3.

40	1.017458
45	0.919491
50	0.813166
55	0.728662
60	0.731694
65	0.576939
70	0.622448
75	0.609539
80	0.654780
85	0.646335
90	0.539000
95	0.530663
100	0.571937
105	0.453086
110	0.476126
115	0.426220
120	0.427976
125	0.478498
130	0.511605
135	0.430662

Writing model shards: 100% 1/1 [00:01<00:00, 1.19s/it]

```
('sft_model/tokenizer_config.json', 'sft_model/tokenizer.json')
```

Questions:

- The formatting function is taking each example from the dataset and then compiling it into a singular text string which follows the specified structure (### Instruction / ### Input / ### Response). The instruction aspect labels the task that the model should be performing, while the input part gives it context or other necessary data needed to complete the task. The response part contains the expected output for the input and instruction combination. It matters due to the fact that it trains the model on the pattern of reading instructions and producing adjacent answers, and the labelling allows the model to distinguish the different pieces of the examples. This allows the model to improve its ability to generalize to new instructions and generate more accurate and relevant responses.

Before vs. After comparison:

```
40    print("AFTER :", run_one(tuned_gen, inst, inp))
41    print("-" * 60)
✓ [10] 19s 753ms

Loading weights: 100%  76/76 [00:00<00:00, 2578.98it/s]
Loading weights: 100%  76/76 [00:00<00:00, 3141.03it/s]
Both `max_new_tokens` (=60) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
 documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
IN: ('Rewrite the sentence to be more polite.', 'Give me that.')
Both `max_new_tokens` (=60) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
 documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Both `max_new_tokens` (=60) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
 documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
BEFORE: I am sorry, but I'm not sure what you're thinking about this question and will try to answer it in a few minutes or so before
 we get there!
AFTER : Could you please give this?
------------------------------------------------------------
IN: ('Classify the sentiment as positive/negative.', 'This is the worst.')
```

```
41    print("-" * 60)
✓ [10] 19s 753ms

Loading weights: 100%  76/76 [00:00<00:00, 2578.98it/s]
Loading weights: 100%  76/76 [00:00<00:00, 3141.03it/s]
IN: ('Classify the sentiment as positive/negative.', 'This is the worst.')
Both `max_new_tokens` (=60) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
 documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
Both `max_new_tokens` (=60) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
 documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
BEFORE: The message should be "I'm not a racist, I am an American." The response to this question was very negative and it's been taken
  down by my parents for being insensitive about race in general. It has also caused me some embarrassment because of how much racism
 they've had on their part since
AFTER : positive
------------------------------------------------------------
IN: ('Extract the city from the text.', 'I drove from LA to San Diego.')
Both `max_new_tokens` (=60) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
```

```
Loading weights: 100%|▰▰▰
```

Questions:
For all three tasks, it is highly noticeable that the structure of the fine-tuned model's responses are much shorter and simpler in length. For many of the responses of the fine tuned model, the model simply outputs one word with the answer or response to the prompt. Contrastingly, the base model's response structure is fairly long and abstract, with a length surpassing a few sentences long. It is also important to note that the base model's responses have little to no relevance to the actual prompt. The base model's responses are very unhelpful and often do not even make sense. On the other hand, the fine-tuned model answers the prompt with relevance, but as for accuracy, it is not always performing well. In the response to both the first and third prompt, the fine-tuned model answered relevant and accurately. As for the second response or the sentiment analysis, the fine-tuned model responded opposing the right answer. Therefore, accuracy is not perfect with the fine-tuned model, but it did often still answer correctly. As for the base model, it never answered relevantly or accurately, and simply generated tokens that were unrelated to the prompt and somewhat unrelated from each other as well.

Task 3 Part C:

Questions:

- The training loss trend is that the loss continues to decrease over time as training continues to proceed. This suggests that the model is learning the patterns better in the dataset as it goes and its accuracy and ability are being improved. As displayed in the loss table, the loss drops from 3.8 at 5 steps in, to 2.6 at 10 steps, then at around 55 to 0.8, then finally at 135 to 0.4. The loss steadily declines and therefore, the loss displays that the model is successfully being trained on the data and therefore performing more accurately and improving its predictions. This also is displayed through the fine-tuned model, which responds much more accurately and relevantly than the general base model before training.

```
Loading weights: 100%|▰▰▰
```

```
✓ [15] 1m 27s
Loading weights: 100%|▰▰▰
```